

1. COMPANY INFORMATION

Company name
SKF

Industry / sector
Machine elements & Production

Company size
Corporate

Website
<https://www.skf.com/se>

Contact person
Meneka Karehalli / Magnus Wahlgard

Role / title
Process Specialist – Machining & Cleanliness
Technology / Digital manufacturing Senior
Strategist

2. THE CHALLENGE**Challenge title**

Analyze and compare learning methods suitable for industrial humanoids

Background & context

Investigate how industrial humanoids can acquire, improve and transfer skills, with emphasis on methods such as manual programming, learning from demonstration and teleoperation, imitation learning, reinforcement learning, simulation-to-real transfer, human feedback and vision-language-action or foundation-model approaches. Teams should compare the methods against industrial requirements including data effort, training time, safety, repeatability, adaptability, computing needs, explainability and deployment maturity. The analysis should distinguish between learning locomotion, manipulation and complete industrial workflows, and identify where hybrid approaches are more realistic than one method alone. A pipeline shall be created to implement the locomotion tasks “Walking up the stairs” or “change of step length” including full documentation

Core question for builders

How can we make the training of humanoids in an industrial setting easier and more efficient?

3A. SCOPE — WHAT IS IN SCOPE**Focus areas / user groups / topics teams should explore**

A structured comparison of learning methods, recommendations for selected industrial task “Vision based Staircase climbing task or step length adjustment” and a practical learning and validation concept

3B. SCOPE — WHAT IS OUT OF SCOPE**Legal, technical, or strategic constraints to be aware of**

- We provide or will consider using Google Clab GPU access for training
(<https://cloud.google.com/products/compute/pricing/compute-optimized>)
- Humanoid and hands data should be provided as input
(https://support.unitree.com/home/en/G1_developer/rl_control_routine)

4. WHAT A GREAT SOLUTION LOOKS LIKE

Success criteria

- Good reasoning and concise conclusions
- Visible effort that the solution was tried to be implemented
- Solid working simulation solution which can be tested in real life
- Business KPIs and values

Deliverable expected

- Try one specific task full pipeline (ends at simulation)
Tasks: Walking up the stairs or setting the step length

Deliverables:

- Results and code
- Functions etc.) or other modalities to train the humanoid
- Dynamic simulation
- Written documentation:
If it works – why?
If it does not work – why?
- A proposed learning pipeline for one SKF use case, including data collection, training, testing and success measures

OR

- Introduction of the training pipeline structure and details of the analysis of at least 2 tasks in the training pipeline (locomotion obligatory and manipulation desired):

Deliverables

- Theoretical investigation of how the pipeline is feasible (especially for manipulation)
- How to implement?
 - pick and put
 - wipe the table with a cloth/clean the whiteboard
- An assessment of benefits, limitations, data needs, safety and industrial readiness
- Written analysis of manipulation and locomotion part
- A proposed learning pipeline for one SKF use case, including data collection, training, testing and success measures

Format preference

Slide deck, documentation and simulation results with videos

5. SUPPORT YOU CAN OFFER TO TEAMS

Data & materials provided

Consultations, data sheets, old simulation results

Reports, datasets, customer research...

Expert availability during event

Virtual support on the first day and upon request from students will be provided

Will someone be on-site to answer questions?

Opportunities after the weekend

Mentorship, internship, thesis partner, pilot

Mentorship, pilot, internship, prize...
